

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA5113 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 50

Annexure A Comparative Analysis

Code of Conduct:

*I **Kartheeka Repalle (192372289)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
Kartheeka Repalle

Annexure A
Comparative Analysis

- 1.** Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec with respect to security services, performance overhead, and suitability for different network scenarios. Analyze which is more appropriate for a VPN deployment.
- 2.** Compare transport mode and tunnel mode in IPSec architecture. Evaluate their effectiveness in securing host-to-host versus network-to-network communication.
- 3.** Perform a comparative analysis of Pretty Good Privacy (PGP) and S/MIME in terms of key management, trust models, scalability, and enterprise usability. Recommend the most suitable approach for a corporate email system.
- 4.** Compare rule-based and machine learning-based email spam detection techniques. Analyze their effectiveness in handling evolving spam patterns and minimizing false positives.
- 5.** Compare manual key management and automated key management (e.g., using IKE) in IPSec. Evaluate their impact on security, scalability, and operational complexity in large organizations.

1. COMPARISON OF AH AND ESP IN IPSEC

ANSWER:

Authentication Header (AH) and Encapsulating Security Payload (ESP) are two major protocols used by IPSec to secure network communication.

Comparison:

Feature	AH	ESP

Confidentiality	No	Yes
Integrity	Yes	Yes
Authentication	Yes	Yes
Encryption	No	Yes
Replay Protection	Yes	Yes
Overhead	Lower	Higher
VPN Suitability	Limited	High

AH provides authentication and integrity of IP packets but does not encrypt the actual data. Therefore, the information remains visible to an attacker monitoring the network.

ESP can provide encryption, integrity, authentication, and replay protection. Because it protects the confidentiality of data, ESP is more commonly used in modern VPN deployments.

Performance:

AH generally has lower processing overhead because it does not perform encryption. ESP requires additional processing for encryption and integrity protection. However, modern VPN devices often provide hardware acceleration, which reduces the performance impact.

Suitability:

AH can be useful when only authentication and integrity are required. ESP is suitable for confidential communication such as banking data, business information, employee traffic, and VPN connections.

For a VPN deployment, ESP is generally more appropriate because VPNs usually require both confidentiality and integrity.

Conclusion:

ESP is preferred over AH for most modern VPN deployments because it provides stronger overall protection while supporting encryption, integrity, authentication, and replay protection.

2. COMPARISON OF TRANSPORT MODE AND TUNNEL MODE IN IPSEC

ANSWER:

IPSec supports two main operating modes: Transport Mode and Tunnel Mode. They differ in the amount of the IP packet that is protected.

Comparison:

Feature	Transport Mode	Tunnel Mode

Protection	Payload	Entire IP packet
New IP Header	No	Yes
Overhead	Lower	Higher
Host-to-Host	Suitable	Possible
Network-to-Network	Less suitable	Highly suitable
VPN Usage	Limited	Common

Transport Mode:

In transport mode, the original IP header remains visible while the payload is protected by IPSec.

Flow:

Host A

|

| Original IP Packet

v

IPSec Protection

|

v

Internet

|

v

IPSec Protection

|

v

Host B

Transport mode is commonly used for direct host-to-host communication.

Advantages:

- Lower overhead.
- Lower processing requirements.
- Efficient for host-to-host communication.

Disadvantage:

The original IP addressing information remains exposed.

Tunnel Mode:

In tunnel mode, the complete original IP packet is protected and encapsulated inside a new IP packet.

Flow:

Network A

|

Gateway A

|

| Encrypted IPSec Tunnel

v

Public Internet

|

v

Gateway B

|

Network B

Tunnel mode is commonly used for site-to-site VPNs and network-to-network communication.

Advantages:

- Protects the complete original IP packet.
- Hides internal source and destination addresses.
- Suitable for site-to-site VPNs.
- Provides stronger network-level privacy.

Disadvantage:

It introduces additional packet overhead.

Conclusion:

Transport mode is more suitable for host-to-host communication, while tunnel mode is more suitable for network-to-network communication. For corporate site-to-site VPNs, tunnel mode is generally preferred.

3. COMPARISON OF PGP AND S/MIME

ANSWER:

PGP and S/MIME are security technologies used to provide secure email communication. Both support encryption and digital signatures, but their trust and key-management models are different.

Comparison:

Feature	PGP	S/MIME

Key Management	User controlled	Certificate based
Trust Model	Web of Trust	CA/PKI
Scalability	Moderate	High
Enterprise Use	Moderate	High
Central Management	Limited	Strong
Integration	Depends on software	Strong
Encryption	Yes	Yes
Digital Signature	Yes	Yes

PGP:

PGP uses public-key cryptography and commonly follows a decentralized trust model. Users can generate and manage their own keys and verify the keys of other users.

Advantages:

- Flexible.
- User-controlled key management.
- Suitable for decentralized environments.
- Strong encryption and digital signatures.

Disadvantages:

- Key management can become difficult.
- Trust relationships can be complicated.
- Less centralized enterprise administration.

S/MIME:

S/MIME uses digital certificates issued by trusted Certificate Authorities. It integrates well with enterprise email systems and centralized PKI.

Advantages:

- Centralized certificate management.
- Strong enterprise integration.
- Easier identity management.
- Suitable for large organizations.
- Supports certificate revocation and renewal.

Disadvantages:

- Requires PKI infrastructure.
- Certificate management is necessary.
- Initial deployment can require additional administration.

Recommendation:

For a corporate email system, S/MIME is generally more suitable because it provides centralized certificate management, structured trust relationships, and good integration with enterprise email systems.

PGP may be preferred when decentralized key management and user-controlled trust are more important.

Conclusion:

S/MIME is generally recommended for large corporate environments, while PGP is useful for users and organizations that prefer a decentralized trust model.

4. RULE-BASED VS MACHINE LEARNING-BASED SPAM DETECTION

ANSWER:

Email spam detection can be implemented using rule-based techniques or machine learning techniques. Both approaches have advantages and limitations.

Comparison:

Feature	Rule-Based	Machine Learning

Method	Fixed rules	Learned patterns
Setup	Simple	More complex
Adaptability	Low	High
New Spam Detection	Limited	Better
Maintenance	Manual	Continuous training
False Positives	Can be high	Can be reduced
Scalability	Moderate	High
Processing	Usually fast	Depends on model

Rule-Based Detection:

Rule-based systems identify spam using predefined conditions.

Examples:

- Suspicious keywords.
- Known spam sender.
- Suspicious domain.

- Too many links.
- Certain attachment types.
- Failed email authentication.

Advantages:

- Easy to understand.
- Easy to implement.
- Fast processing.
- Rules can be manually controlled.

Disadvantages:

- Attackers can modify their messages.
- New spam patterns may bypass existing rules.
- Requires continuous rule updates.
- Excessive rules can cause false positives.

Machine Learning-Based Detection:

Machine learning systems learn patterns from previously classified emails.

Possible features include:

- Sender information.
- Email headers.
- Message content.
- URLs.
- Number of links.
- Attachment information.
- Message structure.
- Previous spam patterns.

Advantages:

- Can identify complex patterns.
- Adapts better to changing spam.
- Can reduce dependence on manually written rules.
- Can process large amounts of email.

Disadvantages:

- Requires training data.
- Requires monitoring and retraining.
- Incorrect training data can reduce accuracy.
- More complex to implement.

Best Approach:

A hybrid approach is often effective.

Flow:

Incoming Email

|

v

Rule-Based Checks

|

v

Header and Reputation Analysis

|

v

Machine Learning Model

|

v

Spam Score

|

+---++

| |

Spam Legitimate

|

Quarantine

Conclusion:

Rule-based detection is simple and fast, while machine learning provides better adaptability to changing spam patterns. A hybrid system combining both approaches can provide stronger detection and help reduce false positives.

5. MANUAL VS AUTOMATED IPSEC KEY MANAGEMENT

ANSWER:

IPSec requires cryptographic keys and Security Associations (SAs) to protect network communication. Key management can be performed manually or automatically using mechanisms such as Internet Key Exchange (IKE).

Comparison:

Feature	Manual Management	Automated IKE

Key Generation	Manual	Automatic
Scalability	Low	High
Administration	Difficult	Easier
Error Risk	High	Lower
Key Renewal	Manual	Automatic
Security	Moderate	Stronger
Large Enterprise	Not suitable	Suitable
Operational Cost	High	Lower

Manual Key Management:

In manual management, administrators configure cryptographic keys and IPSec parameters directly on the devices.

Advantages:

- Simple for small networks.
- Easy to understand.
- No complex key-management infrastructure required.

Disadvantages:

- Difficult to manage many devices.
- Keys must be changed manually.
- High possibility of configuration errors.
- Difficult to revoke compromised keys quickly.
- Poor scalability.

Automated Key Management Using IKE:

IKE automatically negotiates security parameters and establishes cryptographic key material between IPSec peers.

Process:

Gateway A

|

| IKE Negotiation

v

Authentication

|

v

Key Exchange

|

v

Security Association

|

v

Secure IPSec Communication

Advantages:

- Automatic key generation.
- Automatic key renewal.
- Better scalability.
- Reduced configuration errors.
- Easier management of large VPN environments.
- Supports secure authentication and negotiation.

Impact on Security:

Automated key management reduces the risk of old or incorrectly configured keys remaining active. It also allows regular key renewal, improving overall security.

Impact on Scalability:

Manual key management becomes increasingly difficult as the number of VPN devices increases. IKE allows devices to negotiate keys automatically, making it much more suitable for large organizations.

Operational Complexity:

Manual management has a low initial complexity but becomes very difficult to maintain at scale. Automated management requires initial configuration but significantly reduces long-term administrative work.

Conclusion:

For a small test environment, manual key management may be acceptable. For large organizations, automated key management using IKE is strongly preferred because it improves security, scalability, reliability, and operational efficiency.